

CHAPTER 8: DIFFERENTIALS AND PARTIAL DERIVATIVES

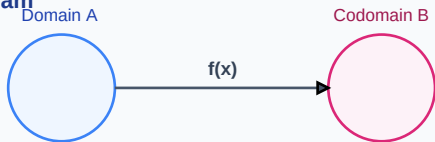
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Mathematics Volume 2
Chapter Name:	Differentials and Partial Derivatives

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Differential	The linear approximation of the change in a dependent variable, denoted $dy = f'(x) \cdot dx$.
Homogeneous Function	A function $f(x,y)$ where $f(tx, ty) = t^n \cdot f(x,y)$ for all real t , where n is the degree of the function.
Partial Derivative	The derivative of a multi-variable function with respect to one variable while holding all other variables constant.

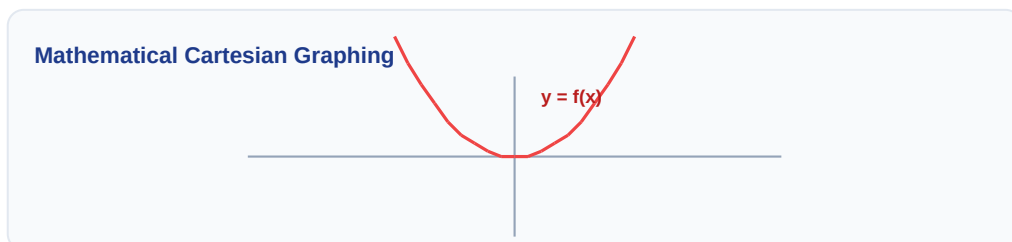
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Linear Approximation: $L(x) = f(x_0) + f'(x_0) * (x - x_0)$
- Euler's Theorem: $x * \frac{df}{dx} + y * \frac{df}{dy} = n * f$
- Total Differential: $df = (\frac{df}{dx}) * dx + (\frac{df}{dy}) * dy$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: If $u = x^2 \cdot y + y^3$, find du/dx and du/dy .

Step-by-step Solution:

$$du/dx = d/dx(x^2 \cdot y) + 0 = 2xy.$$

$$du/dy = x^2 + 3y^2.$$

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Verify Euler's Theorem for $u = \sin^{-1}(x/y)$.

Analytical Solution Strategy:

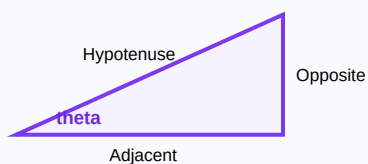
$$u(tx, ty) = \sin^{-1}(tx/ty) = \sin^{-1}(x/y) = t^0 * u. \text{ Degree } n = 0.$$

$$du/dx = (1 / \sqrt{1 - x^2/y^2}) * (1/y) = 1 / \sqrt{y^2 - x^2}.$$

$$du/dy = (1 / \sqrt{1 - x^2/y^2}) * (-x/y^2) = -x / (y * \sqrt{y^2 - x^2}).$$

$$x * du/dx + y * du/dy = x / \sqrt{y^2 - x^2} - x / \sqrt{y^2 - x^2} = 0 = 0 * u. \text{ Verified.}$$

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find differential dy for $y = x^4 - 2x^2$.

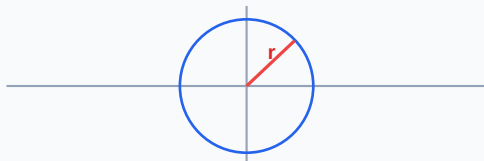
Q2 (2-Mark): Use linear approximation to estimate the value of $\sqrt{26}$. (Take $x_0 = 25$).

Q3 (2-Mark): State Euler's Theorem for homogeneous functions.

Q4 (5-Mark): If $u = \ln(x^2 + y^2)$, prove that $x \frac{du}{dx} + y \frac{du}{dy} = 2$.

Q5 (5-Mark): Evaluate total differential of $z = x^2 \cdot e^y$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Thermal Engineering

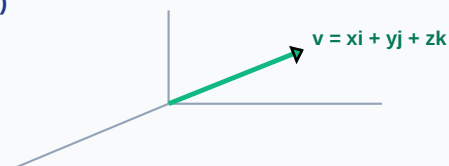
Stress expansion margins in steam pipes are approximated using total differentials.

Application Case: Machine Learning

Backpropagation updates weight parameters via multi-variable chain rule partial derivatives.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

